

Garrett Alexander Kukier

kukier@stanford.edu · (949) 466-8469 · Stanford, CA · ORCID: 0009-0008-4810-7663

EDUCATION

Stanford University — PhD, Chemistry

Sept 2022 – Expected 2027

Advisor: Todd J. Martínez · Advanced to candidacy August 2024 · GPA: 3.82

University of California, Los Angeles — B.S., Chemistry

Sept 2018 – March 2022

GPA: 3.86

PUBLICATIONS

First-author

Kukier, G.A.; Diesendruck, C.; Hait, D.; Xu, R.; Martínez, T.J. "The Force Awakens a Dormant Chemiluminescent Pathway in 1,2-Dioxetane." **J. Am. Chem. Soc.**, 2026.

Kukier, G.A.[‡]; Valenta, L.[‡]; Bhat, V.[‡]; Horst, M.; Martínez, T.J.; Xia, Y. "Diphenyl Naphthocyclobutene as a Dynamic Reversible Mechanophore." *In preparation (May 2026)*. [‡]Co-first author.

Zhou, Q.[‡]; **Kukier, G.A.**[‡]; Gordiy, I.[‡]; Hoffmann, R.; Seeman, J.I.; Houk, K.N. "A 21st Century View of Allowed and Forbidden Electrocyclic Reactions." **J. Org. Chem.**, 2024. [‡]Co-first author.

Kukier, G.A.; Turlik, A.; Xue, X.S.; Houk, K.N. "Violations. How Nature Circumvents the Woodward–Hoffmann Rules and Promotes the Forbidden Conrotatory $4n+2$ Electron Electrocyclization of Prinzbach's Vinylogous Sesquifulvalene." **J. Am. Chem. Soc.**, 2021.

Co-author

Lyu, H.; Tugwell, T.H.; Chen, Z.; **Kukier, G.A.**; Turlik, A.; Wu, Y.; Houk, K.N.; Liu, P.; Dong, G. "Modular synthesis of 1,2-azaborines via ring-opening BN-isostere benzannulation." **Nature Chemistry**, 2024.

Goyal, K.; **Kukier, G.A.**; Chen, X.; Turlik, A.; Houk, K.N.; Sarpong, R. "Rearrangement of a carboxy-substituted spiro[4.4]nonatriene to annulated fulvenes through a Pd(II)-mediated 1,5-vinyl shift." **Chem. Sci.**, 2023.

FELLOWSHIPS & AWARDS

NSF Graduate Research Fellowship (GRFP)

2022 – Present

National Science Foundation

EDGE Doctoral Fellowship

2022 – Present

Enhancing Diversity in Graduate Education — Stanford University

Community Impact Award

April 2026

Stanford Alumni Association

University-wide award recognizing graduate-student community leadership and service.

Art of Science Finalist

May 2025

Stanford Materials Research Society

American Conference on Theoretical Chemistry Best Poster Award

June 2024

Theoretical Chemistry Accounts

ACS Award for Outstanding Senior in Organic Chemistry

June 2022

UCLA Department of Chemistry and Biochemistry

UC LEADS Scholar

2020 – 2022

University of California

Two-year systemwide fellowship; awarded to four UCLA students annually.

Top Honors Presentation Award

March 2021 & 2022

Koret UC LEADS Research and Leadership Symposium

Presentation Award Winner

November 2020

Annual Biomedical Research Conference for Minority Students

RESEARCH EXPERIENCE

PhD Candidate — Department of Chemistry, Stanford University

Jan 2023 – Present

Advisor: Todd J. Martínez

- Mechanistic elucidation for mechanochemistry: discovered a novel chemiluminescent pathway in 1,2-dioxetane, and characterized diphenyl naphthocyclobutene as a dynamic reversible mechanophore.

- Building a random-forest model with engineered quantum-mechanical and RDKit-fingerprint features to predict molecular photostability; collaboration with BASF under the California Research Alliance (CARA).
- Reformulating cavity construction in polarizable-continuum (PCM) solvation models to enable simulation of dissociation phenomena (e.g., NaCl in water) currently inaccessible to implicit solvent — a longstanding open problem with broad implications for solution-phase molecular dynamics.
- Investigating prebiotic reaction networks relevant to the origin of life on Earth and possibly icy moons (Enceladus); using the *ab initio* nanoreactor to test whether metabolic cycles can be rediscovered *a priori* without hypothesis or enzymes.

Undergraduate Researcher — Department of Chemistry and Biochemistry, UCLA

March 2020 – Sept 2022

Advisor: Kendall N. Houk

- DFT-based elucidation of reaction mechanisms and origins of pathway selectivity across electrocyclizations, cycloadditions, and cross-coupling chemistry — first author on two papers (*J. Am. Chem. Soc.*, *J. Org. Chem.*) and contributing author on two additional publications.
- Developed expertise in the Houk-lab distortion–interaction model, orbital symmetry analysis, and multireference methods (e.g., CASSCF) for treating electronically complex systems.

Visiting Undergraduate Researcher — College of Chemistry, UC Berkeley

Summer 2021

Advisor: Richmond Sarpong

- Computational study of a Pd-mediated C–C bond cleavage en route to annulated fulvenes, working in close collaboration with experimentalists to help guide ongoing synthetic efforts (published in *Chem. Sci.*, 2023).

INVITED TALKS & CONFERENCE PRESENTATIONS

“Mechanical Activation of 1,2-Dioxetane Proceeds Through a Novel Chemiluminescent Pathway.” 18th International Congress on Quantum Chemistry (ICQC), Berkeley, CA. *Poster. May–June 2026.*

“Mechanochemical Induction of a New Dissociation Mechanism for 1,2-Dioxetane.” Third Year Seminar, Sapp Auditorium, Stanford University. *Oral. May 2025.*

Juergens, D.; Lu, L.; **Kukier, G.A.** “Predicting Photostability Using Machine Learning and Efficient Quantum Chemistry.” California Research Alliance (CARA) Spring 2025 Program Review, UC Berkeley. *Oral. April 2025.*

“Mechanomechanistic Changes in Dioxetane Dissociation.” American Conference on Theoretical Chemistry, Chapel Hill, NC. *Poster. June 2024.* — *Theoretical Chemistry Accounts Best Poster Award.*

“Mechanomechanistic Changes in Dioxetane Dissociation.” West Coast Theoretical Chemistry Conference, UC Merced. *Poster. May 2024.*

“Dissociation in Implicit Solvent.” West Coast Theoretical Chemistry Meeting, UC Davis. *Poster. May 2023.*

TEACHING, SERVICE & OUTREACH

Founder & Director — Rains Music and Arts Festival, Stanford University

2023 – Present

Founded Stanford’s annual music and arts festival for the graduate school; now a recurring Stanford tradition.

Rains Community Associate — Stanford Graduate Life Office

June 2023 – Present

Organize and host large-scale social and community-building events for the Stanford graduate school.

Instructor — Stanford SPLASH

May 2025

Taught Woodward–Hoffmann orbital symmetry and pericyclic reactions to high-school students from underrepresented backgrounds; students then tested their predictions inside VR with live molecular-dynamics simulations (InterChem).

Temporal Data Sonification Concert — Stanford Center for Computer Research in Music and Acoustics (CCRMA)

March 2025

Public concert presenting dioxetane research through ensemble sonification of molecular-dynamics trajectories; coupled real-time 3D visualization with a live ensemble musical performance of the data.

TECHNICAL SKILLS

Agentic coding tools	Claude Code, ChatGPT Codex, Warp
Programming & ML	Python, C++, PyTorch, shell scripting, Git, LaTeX, HPC / SLURM
Computational chemistry	TeraChem, Q-Chem, Gaussian16, ASE, OpenMM, RDKit, PyMol, ChemDraw
Creative & multimedia	Adobe Creative Suite, Final Cut Pro X, DaVinci Resolve, TouchDesigner, Max/MSP/Jitter
Languages	Spanish (heritage speaker — fluent comprehension, conversational speaking); Japanese (beginner)